7-1-25

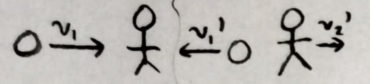
PHY201 Exam on chapter(s): 7, 8

Name

These questions assess your understanding of the material from Chapters 7, 8 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

1. Stephen, a 108-kg ice-hockey goalie who is originally at rest, is hit by a 0.17-kg ice-hockey puck that is traveling at a velocity of 44.0 m/s. Stephen hits the puck in such a way that this may be considered an elastic collision. Calculate the speed of the goalie after he hits the puck.

ANSWER: $0.138\ \frac{m}{s}$ ✓



$$v_2' = \frac{2m_1}{m_1+m_2}v_1 - \frac{m_1-m_2}{m_1+m_2}\cancel{v_2}$$

$$v_2' = \frac{2(0.17\text{ kg})(44\ \frac{m}{s})}{0.17\text{ kg} + 108\text{ kg}} = \boxed{0.138\ \frac{m}{s}}$$

2. James is roller skating, skating over a 4.1-m-tall hill. Assuming that he is moving at a speed of 5.5 m/s at the top of the peak, what is his speed when he reaches level ground at 0 m? Assume that friction is negligible.

ANSWER: $10.52\ \frac{m}{s}$ ✓

$$KE_i + PE_i + W_{nc} = KE_f + PE_f$$

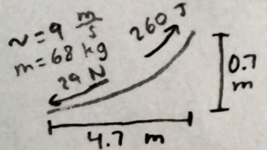
$$\frac{1}{2}mv_i^2 + mgh_i = \frac{1}{2}mv_f^2$$

$$v_f = \sqrt{\frac{\frac{1}{2}mv_i^2 + mgh_i}{\frac{1}{2}m}}$$

$$v_f = \sqrt{\frac{\frac{1}{2}(5.5\ \frac{m}{s})^2 + (9.8\ \frac{m}{s^2})(4.1\text{ m})}{\frac{1}{2}}} = \boxed{10.52\ \frac{m}{s}}$$

3. Gabriel skateboards up a curved ramp (quarterpipe) and launches himself into the air. He begins his motion with a speed of 9.0 m/s at a total distance of 4.7 m away from the launch position. In order to launch himself from the top of the quarterpipe, Gabriel does 260 J of work. Gabriel has a mass of 68 kg and knows that the average friction force acting on him as he skates across this particular surface is 29 N. If the quarterpipe has a height of 0.7 m, how far above the quarterpipe does Gabriel reach?

ANSWER: 3.62 m ✓



$$KE_i + PE_i + W_{nc} = KE_f + PE_f$$

$$W_{nc} = W_G + W_f = W_G + fd\boxed{\cos\theta_{fd}}^{-1} = W_G - fd$$

$$\frac{1}{2}mv_i^2 + W_G - fd = mgh_f$$

$$h_f = \frac{\frac{1}{2}mv_i^2 + W_G - fd}{mg}$$

$$h_f = \frac{\frac{1}{2}(68\text{ kg})(9\ \frac{m}{s})^2 + 260\ \frac{kg\cdot m^2}{s^2} - (29\ \frac{kg\cdot m}{s^2})(4.7\text{ m})}{(68\text{ kg})(9.8\ \frac{m}{s^2})} = 4.318\text{ m}$$

$$h_f = h_q + h_A$$

$$h_A = h_f - h_q = 4.318\text{ m} - 0.7\text{ m} = \boxed{3.62\text{ m}}$$

+3

4. Chloe is riding on a roller coaster, ascending a steep track as she approaches the top of a 80.2-m peak. Assuming Chloe is at rest at the top of the peak, what is her speed when she reaches level ground at 0 m? Assume that friction is negligible.

ANSWER: $39.65 \frac{m}{s}$

$\cancel{KE_i} + PE_i + \cancel{W_{nc}} = KE_f + \cancel{PE_f}$

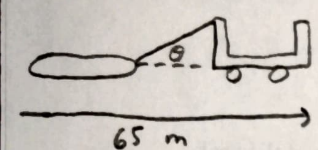
$mgh_i = \frac{1}{2} m v_f^2$

$v_f = \sqrt{\frac{\cancel{m} g h_i}{\frac{1}{2} \cancel{m}}}$

$v_f = \sqrt{\frac{(9.8 \frac{m}{s^2})(80.2\ m)}{\frac{1}{2}}} = \boxed{39.65 \frac{m}{s}}$

5. Alexis drives a 1960-kg tractor that pulls a 540-kg log along the ground for 65 m. The rope connecting the tractor to the log makes an angle of 37° with the ground and is acted on by a tensile force of 8500 N. How much work does Alexis's tractor perform during this motion?

ANSWER: 441246 J

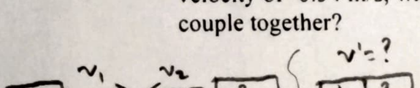


$W = Fd\cos\theta_{Fd}$

$W = (8500 \frac{kg \cdot m}{s^2})(65\ m)\cos(37°) = \boxed{441246\ J}$

6. Train cars are coupled together by being bumped into one another. Suppose two loaded train cars are moving toward one another, the first having a mass of 144000 kg and a velocity of 0.25 m/s, and the second having a mass of 153000 kg and a velocity of -0.54 m/s, where the minus sign indicates the direction of motion. What is the velocity of the two train cars after they couple together?

ANSWER: $-0.157 \frac{m}{s}$



$\vec{p} = \vec{p}'$

$m_1\vec{v}_1 + m_2\vec{v}_2 = m'\vec{v}'$

$\vec{v}' = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m'}$

$\vec{v}' = \frac{(144000\ kg)(0.25 \frac{m}{s}) + (153000\ kg)(-0.54 \frac{m}{s})}{144000\ kg + 153000\ kg} = \boxed{-0.157 \frac{m}{s}}$

T3

7. Joseph places a 0.15-kg toy car next to a spring on a track. He moves the car backward to compress the spring 9.6 cm, then releases the car, propelling it upward along a curved track. How fast is the car going after it is released, before it begins travelling up the slope? Since Joseph greased the axles of the toy car, assume there is no friction. Joseph's previous experiments indicate that the spring has a force constant of 200 N/m.

ANSWER: $3.51 \frac{m}{s}$ ✓

$\cancel{KE_i} + PE_i + \cancel{W_{nc}} = KE_f + \cancel{PE_f}$

$\cancel{\tfrac{1}{2}}k\Delta x^2 = \cancel{\tfrac{1}{2}}mv_f^2$

$v_f = \sqrt{\frac{k\Delta x^2}{m}}$

$v_f = \sqrt{\frac{(200 \frac{kg\cdot m}{s^2\cdot m})(0.096\ m)^2}{0.15\ kg}} = \boxed{3.51 \frac{m}{s}}$

8. Sarah is a 81-kg athlete competing in a race. She begins from rest with an average acceleration of 2.3 m/s² for 4.0 s. What is her momentum at 4.0 s?

ANSWER: $745.2 \frac{kg\cdot m}{s}$ ✓

$\vec{p}_f = m\vec{v}_f$

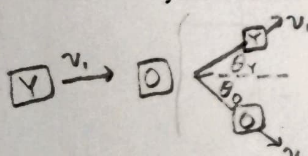
$\vec{v}_f = \cancel{\vec{v}_i} + \vec{a}t$

$\vec{p}_f = m\vec{a}t$

$\vec{p}_f = (81\ kg)(2.3\ \frac{m}{s^2})(4\ s) = \boxed{745.2 \frac{kg\cdot m}{s}}$

9. A 0.14-kg yellow box is slid eastward on a frictionless surface into a dark room, where it strikes an initially stationary orange box, which has a mass of 0.36 kg. The yellow box emerges from the room at an angle of 34° north of east. The speed of the yellow box is originally 8.2 m/s and is 0.8 m/s after the collision. Calculate the speed of the orange box after the collision.

ANSWER: $2.94 \frac{m}{s}$ ✓



$\theta_O = \sin^{-1}\left(\frac{0.174}{v_2'}\right)$

$2.931 = v_2'\cos\left(\sin^{-1}\left(\frac{0.174}{v_2'}\right)\right)$

$v_2' = \boxed{2.94 \frac{m}{s}}$

$\theta_O = 3.397°$

$\vec{p}_x = \vec{p}_x'$

$m_1\vec{v}_{1x} + m_2\cancel{\vec{v}_{2x}} = m_1\vec{v}_{1x}' + m_2\vec{v}_{2x}'$

$m_1v_1 = m_1v_1'\cos\theta_Y + m_2v_2'\cos\theta_O$

$\vec{p}_y = \vec{p}_y'$

$m_1\cancel{\vec{v}_{1y}} + m_2\cancel{\vec{v}_{2y}} = m_1\vec{v}_{1y}' + m_2\vec{v}_{2y}'$

$0 = m_1v_1'\sin\theta_Y - m_2v_2'\sin\theta_O$

$(0.14)(8.2) = (0.14)(0.8)\cos(34°) + (0.36)v_2'\cos\theta_O$

$1.055 = 0.36v_2'\cos\theta_O$

$2.931 = v_2'\cos\theta_O$

$0 = (0.14)(0.8\ \frac{m}{s})\sin(34°) - (0.36)v_2'\sin\theta_O$

$0.36v_2'\sin\theta_O = 0.0626$

$v_2'\sin\theta_O = 0.174$

+3

10. David crashes his car into a tree at 21.1 m/s (about 47 mph) and comes to an abrupt stop in 0.38 s. Find the magnitude of the average force that the seatbelt exerts on David to bring him to a halt. David's mass is 87 kg, which is roughly equivalent to a weight of about 191 lbs.

ANSWER: 4831 N

$\Delta\vec{p} = \vec{F}_{avg} t = \vec{p}' - \vec{p} = m\vec{v}' - m\vec{v}$

$|\vec{F}_{avg}| = \left|\frac{-m\vec{v}}{t}\right|$

$|\vec{F}_{avg}| = \left|\frac{-(87\ kg)(21.1\ \frac{m}{s})}{0.38\ s}\right| = \boxed{4831\ N}$

11. Lauren steps off a platform onto the floor from a height of 3.5 m. If Lauren has a mass of 86.5 kg and lands without bending her knees, her knee joints compress by 0.63 cm. Calculate the force acting on her knee joints. Note that since Lauren steps off the platform, her initial y-velocity is 0 m/s. Considering that 1000 N is roughly equal to 225 lbs, do you think this hurts Lauren's knees?) Yes

ANSWER: 470944 N

$W_{net} = \Delta KE$

$F_L d \cos\theta_{F_L d} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

$\vec{F}_L - \vec{F}_g$

$F_L d = \frac{+1}{2}mv_f^2$

$\vec{v}_f^2 = \vec{v}_i^2 + 2\vec{a}\Delta\vec{y}$

$v_f = \sqrt{2(-a)\Delta y}$

$F_L = \frac{\frac{1}{2}m(2(-a)\Delta y)}{d}$

$F_L = \frac{\frac{1}{2}(86.5\ kg)(2(9.8\ \frac{m}{s^2})(3.5\ m))}{0.0063\ m} = \boxed{470944\ N}$

12. Ashley normally requires 13500 kJ (about 3226 Calories) of food energy per day. If Ashley consumes 14310 kJ per day, she will steadily gain weight. How much time must Ashley spend bicycling per day, if she does not want to gain weight? Ashley's average power output while bicycling is 440 W. Report your answer in minutes.

ANSWER: 30.68 min

$P = \frac{W}{t}$

$t = \frac{W}{P}$

$t = \frac{14310000\ \frac{kg \cdot m^2}{s^2} - 13500000\ \frac{kg \cdot m^2}{s^2}}{440\ \frac{kg \cdot m^2}{s^3}} = 1840.909\ s$

$\frac{1840.909\ s}{} \Big| \frac{1\ min}{60\ s} = \boxed{30.68\ min}$